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The paleomagnetism and U-Th dating of Mexican stalagmite, DAS2

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A 72 cm high stalagmite from Mexico, DAS2, has been analyzed for its paleomagnetism, and has provided a record of secular variation beginning about 1200 years ago to present. The stalagmite did not grow uniformly, and a low content of magnetic material has rendered the record less precise in some parts. A topographic test involving the magnetic measurement of coeval specimens from the sides of the stalagmite has shown that the record is detectably free from depositional errors. Age data were provided by the U-Th method, and concordancy checked by ^{14}C and ^{226}Ra methods. Shallow inclinations at the base of the record are followed by a partial anticlockwise loop in the declination-inclination plot. There is good agreement between the DAS2 data and the southwest U.S.A. archeomagnetic record and with a lacustrine record from Oregon, from 750 to 1400 AD.

Comparison of the record with the present field at the site latitude shows that westward drift of the field has persisted for at least 1200 years at a drift rate of about 0.30° yr^{-1}. A close match in declinations with the present field at the site latitude suggests that the stalagmite may have recorded the passing of the northern edge of the Gulf of Guinea negative anomaly about 1000 years ago, though the corresponding inclination feature is not present in the record. An archeomagnetic synthesis by Merrill and McElhinny of worldwide data to estimate positions of the main dipole shows that it was tilted away from Mexico from 500 to 1250 AD, and this may account for the shallow inclinations seen at the start of the record.

1. Introduction

Approximately 80% of the internal magnetic field of the earth is, at present, of dipolar form. The hypothetical geocentric dipole modelled from this part is inclined at about 11.5° to the spin axis and is changing slowly in magnitude and direction [1]. The remaining 20% of the field is referred to as the non-dipole field. This part is mainly characterized by large regional anomalies which are growing, decaying or changing shape, and most of which drift westward with respect to the solid earth. By several methods Bullard et al. [2] showed from observatory data, that westward drift was characteristic of most of the non-dipole field. For example an anomaly in the Gulf of Guinea, having a large negative vertical component, was shown to have drifted westwards at a rate of about 0.28° yr^{-1} from 1907.5 to 1945.

Runcorn [3] and Skiles [4] showed that a hypothetical westward-drifting radial dipole below the core-mantle boundary will produce clockwise looping of secular variation (SV) in the northern hemisphere, though there may be exceptions to this rule [5]. The converse of this rule, that is that clockwise (anticlockwise) looping in the field direction is solely due to westward (eastward) drifting core sources may not necessarily be true [6]. Growing or decaying sources, together with movement of the main dipole may produce SV loops of either sense (e.g. [7,8]).

It is natural for paleomagnetists to look for patterns in the SV (cyclic declination, D, and inclination, I features, dipole wobble or oscillation, periodicity) in order to characterize the core mechanisms that cause SV. More recent discussions of sediment paleomagnetic records have either tended to place less emphasis on cyclic SV, or have pointed out that recurring D or I features have variable periods (e.g. [9]). In the earlier papers of Creer and co-workers (see e.g. [10,11]), they showed that fixed radial dipoles, oscillating in magnitude, might also account for cyclic D and I features. This is consistent with the analysis of observatory data by Yukutake and Tachinaka [12] and Yukutake [13] who showed that the modern

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non-dipole field consisted of both drifting and standing parts. The task of interpretation will also be made more difficult if the main dipole field has changed in magnitude and direction at about the same rate as the non-dipole field. From the analysis of mainly archeomagnetic data, changes in the intensity of the main dipole appear to have been quite slow and the ratio of the non-dipole to the dipole field intensity has been about the same over the past 50 ka [14].

In this paper we discuss the global circulation of the Gulf of Guinea anomaly, and its leading edge of westerly declinations. The paleomagnetic record comes from a stalagmite, DAS2, from Mexico, and covers the last 1200 years. The ages have been assigned by ^{230}Th-^{234}U, ^{14}C and ^{226}Ra analyses, and by the observation that the stalagmite top is modern. These and other data discussed herein, suggest that the wobble of the main dipole may have considerable effect on both the looping and bias in the field direction.

2. Field methods and sample preparation

Stalagmite DAS2 was taken from the cave Sotano del Arroyo, near Valles, Mexico, latitude 22.09°N, longitude 99.0°W. The sample was oriented with respect to azimuth using an aluminum device carrying a Suunto magnetic compass, and a correction for local declination was applied to the compass readings. The horizontal plane was determined by water levels in a three-arm clear plastic U-tube. Though the top of the stalagmite was dry it was also fresh, and we believe that it was actively forming. The sample was separated from a calcite substrate by hammer and chisel.

In the laboratory the stalagmite was divided into four subsections for ease of cutting, and each of these was cast in plaster of Paris so that orien-

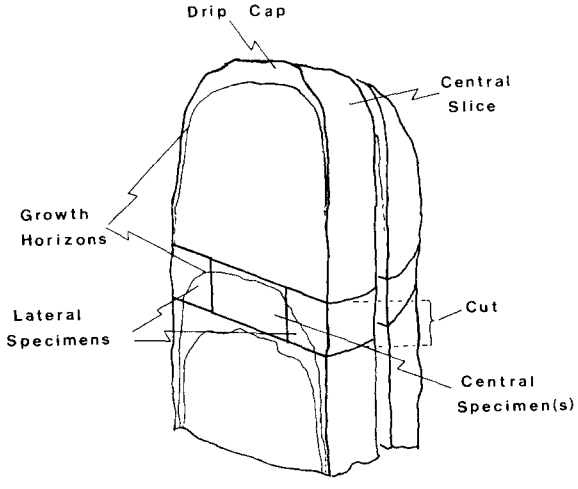


Fig. 1. Cartoon to show growth morphology and the notation used for specimens prepared from a stalagmite.

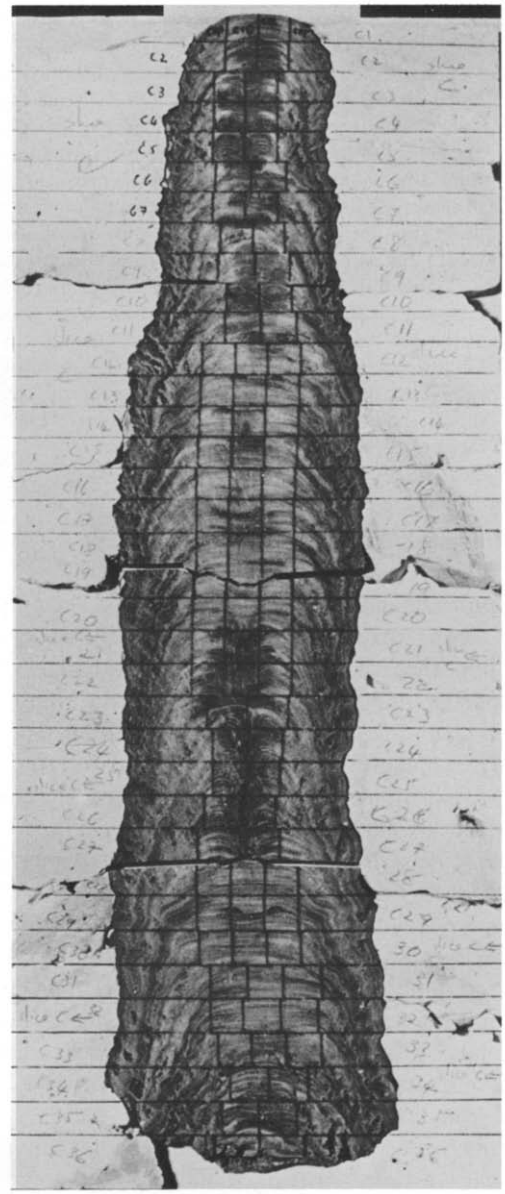


Fig. 2. Photograph of a central slice of DAS2, and the markings prior to making the cuts.

tation marks were parallel to the sides of the mould. Fig. 1 shows how the saw cuts were made for specimen preparation, with the accompanying notation. Central and lateral specimens were usually about 2 cm thick. The cut numbers are labelled from the top, No. 1, downwards. Two slices were taken from the centre of DAS2 yielding 4, 5 or 6 central specimens per cut.

3. Description of the stalagmite

Fig. 2 is a photograph of a central slice of this stalagmite showing how the diameter varied. The smaller diameters, from 55 to 37 cm from the top, and 20 cm up to the top corresponded to an increase in the thickness of the growth layering and this suggests that the stalagmite grew more rapidly over these zones. The U-Th data are not good enough to detect this (sections 6 and 7).

As the stalagmite grew, floods coated the surface principally with mud. Most of it was subsequently washed off the drip-cap area by the splash of the drip water. Where the diameter is smaller and the growth layers are thicker the detritus is less prominent. Thus it appears that over these zones the drip rate had increased so as to accelerate the growth and, perhaps, to more thoroughly wash off or dilute the detrital material. There were a few

TABLE 1

AF-cleaned magnetization directions and VGPs of combined cuts for DAS2

Cuts	N	Direction		k	δD	δI (α_{95})	VGPs		Age (ka)
		D	I				lat.	long.	
1, 2	8	11.5	38.5	208	4.4	3.4	79.0	351.5	45
2, 4, 6	12	14.0	33.0	63	6.1	5.1	76.5	6.5	90
4, 6, 8	11	15.0	31.0	76	5.7	4.8	75.0	9.0	130
6, 8, 10	10	14.0	32.5	75	6.0	5.0	76.0	6.5	175
8, 10, 11	10	10.5	33.0	221	3.6	3.0	79.5	11.0	220
10, 11, 12	13	6.5	31.5	162	3.6	3.0	82.0	29.0	260
11, 12	10	4.5	30.5	245	3.3	2.8	83.0	44.0	310
13	6	0.0	28.5	726	2.4	2.1	83.0	82.5	350
14	6	1.0	27.0	646	2.5	2.2	82.0	74.0	395
15	5	0.0	28.5	333	3.9	3.4	83.0	81.5	435
16, 17	10	358.5	32.0	278	3.1	2.7	85.0	99.5	480
18	6	1.5	30.0	454	3.1	2.7	84.0	68.5	525
19, 20	9	355.5	33.5	187	4.1	3.4	84.5	128.5	570
20, 21, 22	15	358.0	33.0	123	3.9	3.3	85.5	108.0	610
21, 22, 23	14	358.5	33.5	105	4.4	3.6	86.0	103.5	660
22, 23, 24	11	356.5	32.5	123	4.5	3.8	84.5	119.5	700
23, 24, 25	10	356.5	31.5	98	5.2	4.5	84.0	116.5	745
24, 25, 27	10	354.0	32.5	125	4.7	4.0	83.0	133.5	790
27	4	352.0	34.0	408	4.2	3.5	81.5	148.0	830
28	6	346.5	33.0	623	2.7	2.3	77.0	155.5	875
29	6	344.0	33.0	403	3.4	2.8	74.5	158.5	920
30	6	343.0	34.0	534	3.0	2.5	73.5	162.5	960
31	6	340.5	33.5	547	2.9	2.4	71.5	162.5	1005
32	6	342.0	33.0	3068	1.2	1.0	72.5	160.5	1050
33	6	349.5	30.5	1036	2.1	1.8	78.5	143.5	1095
34	4	353.0	33.0	423	4.1	3.4	82.5	139.0	1135
35	4	356.5	24.5	333	4.2	3.8	80.0	102.5	1180
36	2	356.5	22.5	223	7.2	6.6	79.0	100.0	1224
Lateral specimens, thermally demagnetized to 200°C									
C30P		348.5	36.0						
C30T		346.5	32.0						

Notes: Data used in Figs. 5, 6a, 7 and 8. N is the number of specimens for that group of cuts. k is Fisher's precision estimate. α_{95} is the radius of the circle of confidence at the 95% level. Ages were recalculated by taking a basal age of 1224 years and dividing by the position of the data point.

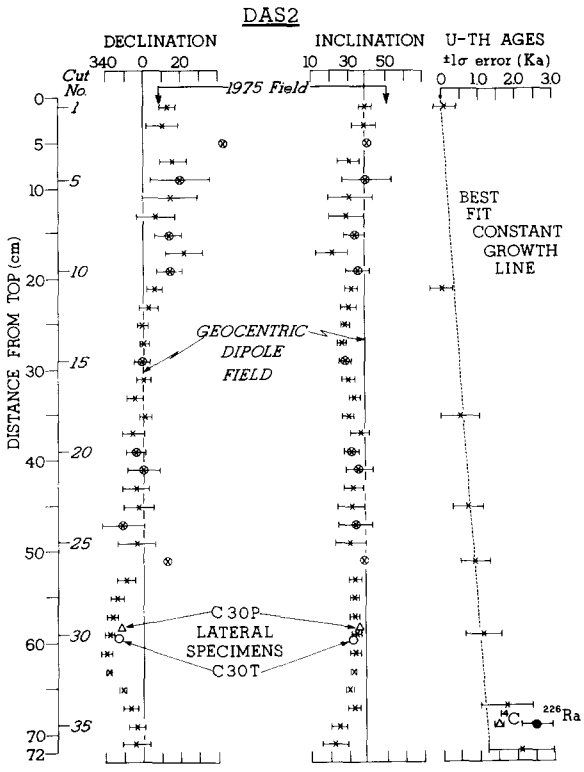


Fig. 3. The AF-cleaned declinations and inclinations, the corrected ^{230}Th-^{234}U, ^{14}C and ^{226}Ra dates of DAS2 against distance from the top. The magnetizations are from Appendix 1. Circled crosses denote that one or more specimen directions have been omitted to obtain the second mean directions (see text). The circled crosses without error bars indicate that they have not been used in subsequent analyses. The D, I error bars are based on α_{95}. The points for lateral specimens C30P and C30T have been plotted above and below their actual position for clarity.

euhedral opaque grains observable in thin section, up to 25 μm in length, which seemed to have been etched before becoming incorporated into the calcite fabric. In other speleothems examined by us [15,16] opaque grains occurred along growth planes together with organic material, quartz and other non-calcite grains. It is clear that those opaque grains that are microscopically observable were part of the flood load. This conclusion is almost certainly valid for DAS2 despite the scarcity of observable opaque grains. In reflected light they showed a uniform grey colour suggestive of magnetite. Curie temperatures from two lateral specimens were estimated to be between 540 and 600°C. Saturation of isothermally induced remanence of a third lateral specimen occurred at about 2 kOe (200 mT). These observations support the suggestion that magnetite was the carrier of the NRM, and essentially exclude the presence of hematite. It is not certain to us whether the magnetization is of detrital or chemical origin, or a mixture of both. We intend to address this question generally in a separate paper.

In order to test whether the (possible) detrital magnetization was the cause of any depositional errors, two lateral specimens were measured using the magnetometer. If depositional errors existed then it might be expected that their magnetization directions would be somewhat different from those of specimens from the corresponding central drip areas. The specimens were chosen for their high detrital content. They were C30P and C30T, taken from opposite sides of slice C of cut 30. Their data, though an average over several layers (Table 1 and Fig. 3), show that the NRMs and thermally cleaned directions were in accord with each other and with the directions from the central specimens. It is therefore concluded that there are no significant directional errors due to the deposition of ready-magnetized grains from the flood-laid detritus.

4. Measurement of the natural and cleaned remanent magnetizations of DAS2

The NRM of each specimen was measured using a double axis SQUID magnetometer having a noise level of about 2×10^{-7} Oe ($\equiv 1.6 \times 10^{-5}$ A m^{-1}). One central specimen from cut 32 was used as a pilot for alternating field (AF) demagnetization. The highest peak field was 1 kOe (100 mT). Fig. 4a shows this demagnetization in terms of the ratio of the moment M at each AF step to the natural moment M_0. The median destructive field was 255 Oe (25.5 mT). Up to the 100 Oe (10 mT) step there was movement of the vector away from the modern field direction after which the direction was stable (Fig. 4b). We thus infer that the low coercive force part of the NRM was a viscous remanent magnetization (VRM) acquired in the modern field. It is probable that any movement of the direction after about the 500 Oe (50 mT) step was due to small anhysteretic remanent magnetizations (ARMs), acquired via the magnetically softer grains. The stable remanence is considered to be primary and that it formed when the growth layer formed.

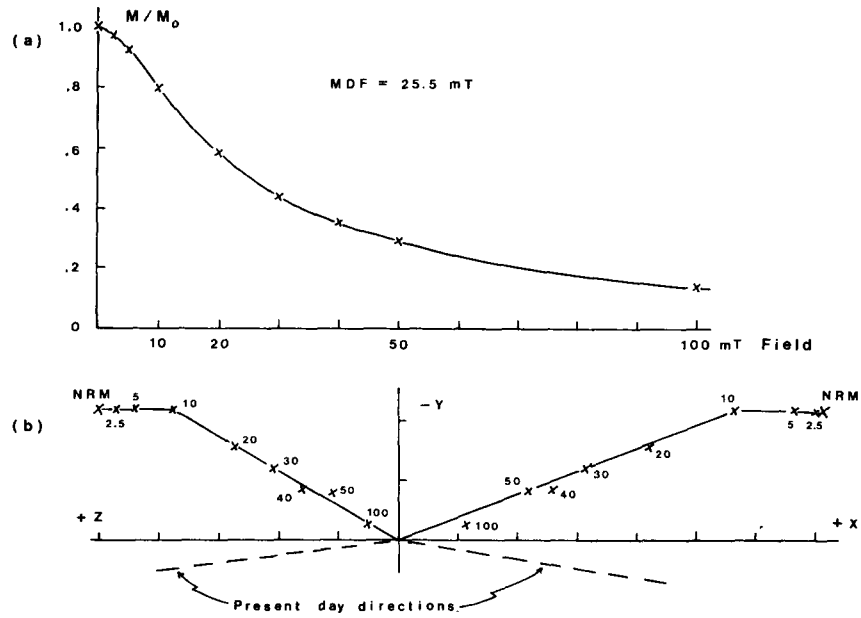


Fig. 4. (a) The alternating field (AF) demagnetization of specimen C32S as the ratio of the moment M at the AF step to the initial, natural magnetization M_0. (b) The corresponding Zijderveld plot of specimen C32S. X, Y and Z are the site components of its moment and are compared with the present-day (1975) field directions. The numbers next to the points are the AF steps in mT.

Unfortunately many specimens had low intensity NRMs. For example, specimens from cuts 3, 4 and 5 had moments of from 1×10^{-7} Oe g^{-1} (1×10^{-7} A m^2 kg^{-1}) down to 0.13×10^{-7} Oe g^{-1} (0.13×10^{-7} A m^2 kg^{-1}). Specimens of cuts 23–25 had low intensities and those for cut 26 were especially low ($\approx 0.1 \times 10^{-7}$ Oe g^{-1}, $\approx 0.1 \times 10^{-7}$ A m^2 kg^{-1}) Because of these low intensities a general cleaning AF of 55 Oe (5.5 mT) was chosen as a compromise between adequately removing the low coercivity VRMs and leaving a sufficiently measurable magnetization above the noise level of the magnetometer. Fig. 4b indicates that the 55 Oe (5.5 mT) AF step will not have removed all of the VRM of some specimens.

5. The paleomagnetism of DAS2

The NRMs, AF-cleaned directions and virtual geomagnetic pole (VGP) positions of the DAS2 record are presented in Appendix 1 and Table 1 as follows:

The specimen directions for each cut have been combined to derive a mean as though they came from a random distribution of directions about the mean, using Fisher statistics. This is not valid in some cases because the original data show that directions of specimens incorporating the curved parts of the growth layers lay between the directions of central specimens and the direction of the succeeding upper cut (original data not shown for brevity). This has probably not affected the mean of the cut by much, but the distribution about the mean is probably not a circle, but an ellipse elongate in the direction of the SV.

Where the AF-cleaned direction of a specimen has not fallen within the α_{95} of the mean of the cut it has been omitted in order to obtain a second estimate. The precision of the second mean has improved considerably for most cuts treated this way (Appendix 1 and Fig. 3).

Measurement of the magnetic remanences of specimens from the "fast growth" layers (see section 4) showed: (1) generally weak signal strength, and (2) where the signal strength is adequate, a tendency to cluster in direction with adjacent layers, and (3) where the signal strength is very weak, a tendency to scatter in direction so as to destroy any serial correlation with neighbouring segments of the D-I track. The means of these "fast-growth" cuts have also been combined, therefore, using a 2- or 3-point moving window to

obtain a smoother SV curve (Table 1 and Figs. 5, 6a, 7–9). The layers concerned are represented by cuts 26 to 18 and cuts 10 to 1. The estimated ages of some of the data points have also changed as a result of this treatment (Table 1).

The absence of detectable depositional errors and the absolute orientation data have enabled us the produce the VGPs of Table 1, and to plot the VGP path of Fig. 6a.

The fresh appearance of the top and the age data (section 7) show that the top is modern. The magnetization direction of cut 1 and 2 combined, would of course, be somewhat older than present day values. In Figs. 5, 6a, 8 and 9 the path segment from cuts 1 and 2 to the 1975 direction or VGP has been completed with a dashed-dot line.

6. The uranium-thorium dating of stalagmites

Stalagmites may be age dated by ^{230}Th-^{234}U method [15], which relies on the following observations. Uranium grandparent ^{238}U and parent ^{234}U are transportable in groundwaters as dissolved carbonate ions whereas daughter ^{230}Th hydrolyzes and readily precipitates onto solid surfaces. Thus as a stalagmite layer is deposited from the feedwater, uranium is incorporated into the calcite whereas ^{230}Th is absent. Subsequently ^{230}Th, with a half-life of 7.52×10^3 years grows back towards equilibrium with parent ^{234}U. The α-particle activity ratio ^{230}Th/^{234}U is a measure of the return to equilibrium and hence of the age. The dating limit of the method is about 350 ka.

In sample DAS2, past cave floods have deposited detritus onto its growth surfaces and have thereby probably introduced some common ^{230}Th. Its presence is inferred from the presence of ^{232}Th in the thorium α-spectra. If present, this allogenic ^{230}Th is indistinguishable from the radiogenic ^{230}Th, and thus causes an apparent increase in the age. In the age-data presented here, age corrections have been made to the raw data by using an assumed initial ^{230}Th/^{232}Th ratio. For details of this type of correction see [16].

7. The age-dating of DAS2

The U-Th activity ratios, the U and Th concentrations and the computed ages of DAS2 are given in Table 2.

Because of the low age of the stalagmite the ^{230}Th/^{232}Th activity ratios are low. This has resulted in fairly large shifts in the corrected ages. Such large shifts are more susceptible to error if the value that is chosen for the initial ^{230}Th/^{232}Th ratio is wrong. The value for $(^{230}\text{Th}/^{232}\text{Th})_0$ that we have used is 1.5, and is based on the initial ratio of 1.8 ± 0.6 for cut 1, 1.5 ± 0.5 for cut 11, and on the estimates given in [17–19]. The age of cut 34 was shifted from 3.3 to 1.7 ka and the basal age was shifted from 6.5 to 2.1 ka. These corrected ages are in agreement with the growth trend from the other six ages alone, though they are slightly older than is predicted by the constant growth line.

Only the simplest assumption of a constant growth rate is warranted by the corrected age data. In obtaining a linear least squares fit it has also been assumed that the top is modern. That is, the least squares fit utilized nine data points including a zero intercept on the age versus length plot. The inverse of the variance in these ages has been used as a weighting factor in the fit. Cut 11 is out of stratigraphic order with the other data points. The one standard deviation of cut 11 falls on the constant growth line at about the 75% confidence level; it has been retained in the growth rate estimate. The linear least squares fit to the nine data points has the following properties (Fig. 3):

(1) A reciprocal growth rate of 0.017 ± 0.004 kyr cm^{-1}, which corresponds to an average growth rate of 0.059 cm yr^{-1}.

(2) The age of cut 36, the base is 1224 a, and therefore the growth interval is 1224 yr.

As tests for concordancy, we have also analyzed specimens from the top and near the base to produce ^{14}C and ^{226}Ra ages (Table 3). In the case of the ^{14}C ages (produced at Simon Frazer University, British Columbia), we have assumed that between the base and the top of the stalagmite, (1) the same fraction of "dead" to organic carbon was present, and (2) the chemical fractionation of ^{14}C between the dripwater and the calcite was the same. The ^{14}C ages are 1060 ± 100 yr BP for the top and 2510 ± 100 yr BP for cut 35. Thus since the top is modern, and using the 5730 year half-life, the age of cut 35 is 1490 ± 140 ka.

^{226}Ra ages were obtained by the scintillation-counting of the ^{226}Ra daughter, of specimens dis-

TABLE 2

Uranium-thorium activity ratios, uncorrected and corrected ages of DAS2

Cut No.	Distance from top (cm)	U conc. (ppm)	$^{230}Th/^{234}U$	$^{230}U/^{238}U$	$^{230}Th/^{232}Th$	Age $\pm 1\sigma$ error (ka)	
						uncorrected	corrected
1	1.2	0.70	0.007 ± 0.001	1.08 ± 0.02	1.8 ± 0.6	0.8 ± 0.1	0.1 ± 0.3
11	31.1	0.57	0.006 ± 0.001	1.11 ± 0.02	1.5 ± 0.5	0.7 ± 0.1	0.0 ± 0.3
18	35.2	0.65	0.015 ± 0.002	1.08 ± 0.03	1.2 ± 0.6	1.6 ± 0.2	0.5 ± 0.5
23	45.1	0.59	0.009 ± 0.002	1.07 ± 0.02	4.8 ± 3.3	1.0 ± 0.2	0.7 ± 0.4
26	51.1	0.42	0.013 ± 0.002	1.06 ± 0.02	3.6 ± 1.4	1.5 ± 0.2	0.9 ± 0.4
30	59.1	0.58	0.026 ± 0.002	1.075 ± 0.02	2.4 ± 0.4	2.8 ± 0.2	1.1 ± 0.5
34	67.1	0.54	0.030 ± 0.003	1.09 ± 0.03	3.1 ± 0.8	3.3 ± 0.8	1.7 ± 0.7
36	72.0	0.43	0.058 ± 0.004	1.07 ± 0.02	2.2 ± 0.3	6.5 ± 0.4	2.1 ± 0.9

solved in 9*N* HCl. ^{226}Ra, with a half-life of 1602 years, is precipitated from the dripwater and substitutes for Ca in the calcite. For such a young stalagmite it is safe to assume that any ^{226}Ra produced from ^{230}Th, in situ, is negligible. In producing an age for cut 35 we have also assumed, less safely, that its specific activity at the time of deposition was the same as that of the top. The age of cut 35 was then computed from $t = (1602/\ln 2) \cdot \ln(A_{top}/A_{35})$, where A_{top} and A_{35} are the specific activities of cuts 1 and 35. The radium age of cut 35 is 2470 ± 390 ka.

These two ages have been plotted in Fig. 3 with their 1σ counting errors. The ^{14}C age is in good agreement with the age of cut 35 deduced from the best fit line from the ^{230}Th ages. The radium age is more in agreement with the ^{230}Th ages of cuts 34 and 36 considered separately. There is thus a moderate possibility that the basal layers of the sample are as old as 2 ka.

TABLE 3

^{14}C and ^{226}Ra age data for DAS2

Specimen	pMC [a]	^{14}C age	Corrected age $\pm 1\sigma$ (a) [b]
C2P [c] (SFU376) [d]	87.3 ± 1.3	1060 ± 100	0
B35R (SFU375) [d]	72.9 ± 1.3	2510 ± 100	1490 ± 140

Specimen	^{226}Ra activity (dpm g^{-1})	Age $\pm 1\sigma$ (a)
C2T [c]	0.169 ± 0.011	0
C35R	0.058 ± 0.009	2470 ± 390

[a] pMC denotes absolute "percent modern carbon", as defined by Stuiver and Polach [27].

[b] Age-corrected using the 5730 year half-life (see text).

[c] Cut 2 lateral specimens contained the uppermost (youngest) growth layers.

[d] Simon Fraser University reference.

8. Geophysical interpretation of the DAS2 record

8.1. Features of the record

From Figs. 3, 5, 6a and 7 the following features stand out:

(1) The inclinations are consistently shallower than the inclination expected for an axial geocentric dipole, I_{agd}. I_{agd} is 39.1° for this latitude, whereas the inclinations for most of the record are about 32°.

(2) Most of the field change is registered in declination.

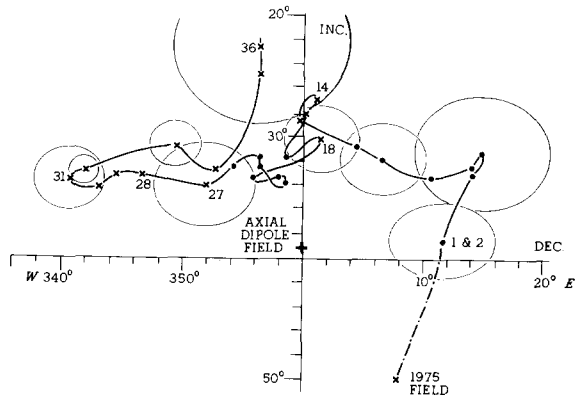


Fig. 5. The Bauer plot of DAS2 from Table 1, with typical errors, based on α_{95}. The numbers are the cut numbers. Crosses are single cuts, heavy dots are window-averaged directions; this notation is also used in the figures that follow.

In detail the field changes are, starting from the base:

(3) Far-sided VGPs describe a left-handed path.

(4) This is followed by abrupt change and a long swing back to a pole position opposite the site longitude, thus producing a hairpin loop.

(5) There appears to be at least one, and possibly two, pauses in SV, followed by,

(6) A resumption of right-handed movement (seen from the observer's longitude) and slowly steepening inclinations (less far-sided VGPs).

(7) Near the top of the record there may be another pause in SV (though this apparently low SV is probably due to poorer quality data).

(8) Completion of the path to the modern-day VGP involves near-sided movement, corresponding to steepening inclinations. However, the inclinations of cuts 1 and 2 are shallower than those of the last 100 years by about 10°.

8.2. *Comparison with archeomagnetic and lacustrine data*

Fig. 6a and b shows the comparison of the DAS2 VGPs with ^{14}C-dated VGPs from the southwest U.S.A. from archeomagnetic data, as averaged by Sternberg [20]. Fig. 6b shows a left-handed hairpin polar track similar to, but not as pronounced as in the DAS2 record, and the path crosses over itself in a clockwise loop. The appearance of all the south and southwest U.S.A. data as compiled by Sternberg (his figure 4-43) is very similar to the lower half of the DAS2 record, though the age data of the archeomagnetic record over the left-handed loop are about 120–170 years younger. Verosub and Mehringer [21] have compared the archeomagnetic data with ^{14}C dated cores from Fish Lake, Oregon, where the same geomagnetic feature was also clearly recorded. Their dates are younger by 80 years, generally, than for the DAS2 data. Part of these age differences may very well be due to the westward drift since these sites lie to the west of the DAS2 longitude.

One notable difference in DAS2 compared to the other two records is a shallow average inclination, which may be due to the latitude difference between the sites with respect to the position of the source of the feature. Considering the paleomagnetic and dating errors that are involved,

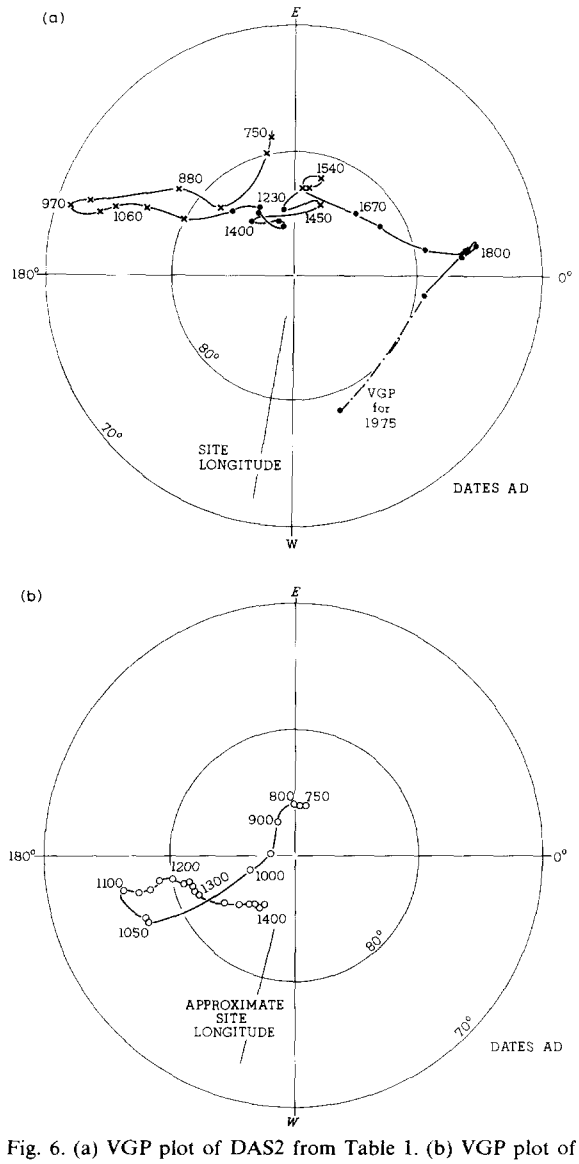


Fig. 6. (a) VGP plot of DAS2 from Table 1. (b) VGP plot of southwestern U.S.A. archeomagnetic data from the compilation of Sternberg [20].

the three types of record (involving two different dating methods) show very good agreement.

8.3. *Comparison with "westward-drifted fields" and the main dipole field*

Fig. 7a and b show comparison of the DAS2 record with; constructions of westward-drifted D and I values taken from

(1) the 1975 IGRF (World Magnetic Charts,

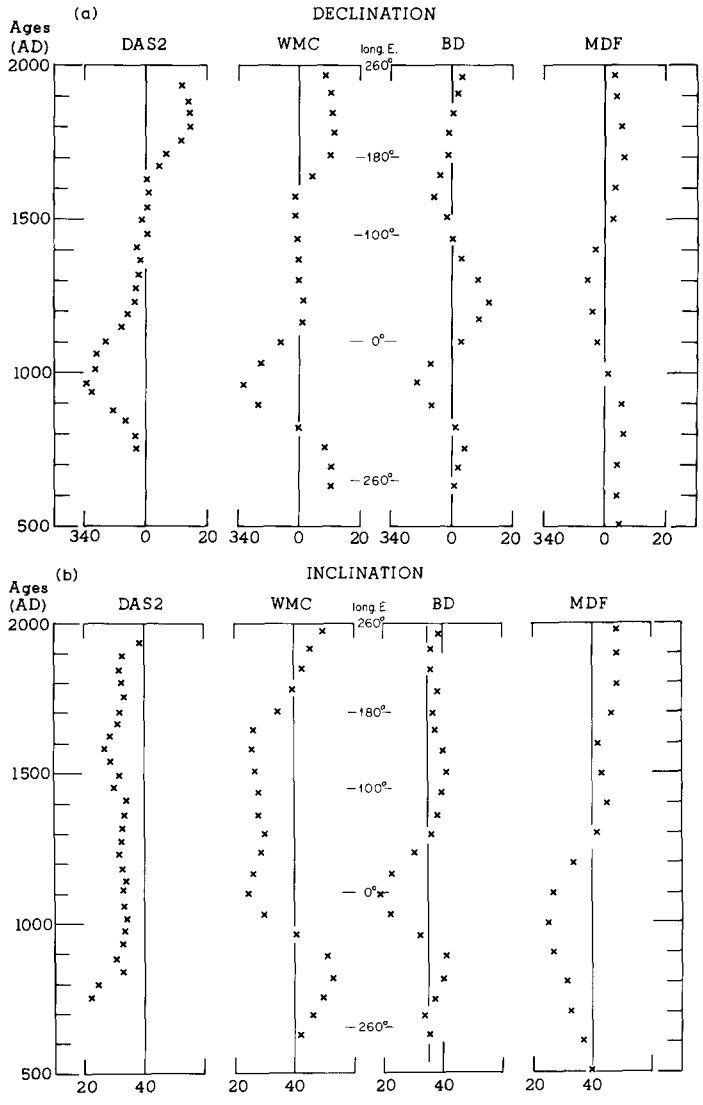


Fig. 7. Comparison of (a) declination and (b) inclination of DAS2 with those of; International Geomagnetic Reference Field for 1975 (WMC) and of Bullard et al. [2] (BD), as though they had drifted westward past the site; the main dipole VGP estimates of Merrill and McElhinny [22] (MDF). The variations of WMC and BD are matched with DAS2 at the maximum of westerly declinations.

WMC) at latitude 20° north and for which the main dipole remains tilted at 11.5° in the direction of Mexico.

(2) The non-dipole values of Bullard et al. [2] (referred to as BD) at 20° north, together with an axial geocentric dipole having the intensity of 1945, as quoted by them. (The differences in D and I between 1945 and 1975 are about comparable with the best precision of a few of the means in the DAS2 record, and do not significantly affect the comparison.)

(3) the field at the site expected from the main dipole (MDF), as 100 year averages, from the archeomagnetic compilation of Merrill and McElhinny [22]

8.4. Westward Drift and the Gulf of Guinea Anomaly

The topmost D and I values of the WMC and BD constructions are the present day ones at 260°E (100°W) (Figs. 7 and 8). The time scale of

their variations has been adjusted by matching the westerly declination maximum of the three sets of data. This declination maximum is now situated in the Atlantic region at about 40°W and is moving westwards at about 0.28° yr^{-1} [2]. The leading edge of 0° declination, which corresponds to the start of the DAS2 record (within our quoted error) is now at about 80°W (1975). Thus DAS2 appears to have recorded almost one complete cycle of the westward drift of all declinations presently at this latitude.

If as seems likely, the declination feature was spatio-temporal (as opposed to pulsating), then we may estimate a drift-rate for it. Taking the westerly declination maximum to be 1005 years old (Table 1), then the mean drift rate of declinations was about $0.30 \pm 0.07°$ yr^{-1}. If we use the archeomagnetic age data (long., 110°W; west. D_{max}, 875 ± 50 years old) then it was $0.33 \pm 0.02°$ yr^{-1}, and for the Fish Lake data (long., 118°W; west. D_{max}, 925 ± 50 years old) it was $0.30 \pm 0.02°$ yr^{-1}.

Part of the westerly declination in the Atlantic at present is due to the tilt of the main dipole of 11.5° at 70°W. The tilt alone would produce maximum westerly declinations of 12° at about 160°W. The difference that the tilt makes to both declination and inclination may be seen by comparing the WMC and BD constructions. Thus the shallow inclinations in WMC compared to BD from 40° to 140°E are mostly due to the tilt of the main dipole away from these longitudes. The steeper inclinations in WMC from 260°E (100°W) to 290°E (70°W) are due to the tilt of the dipole toward these longitudes.

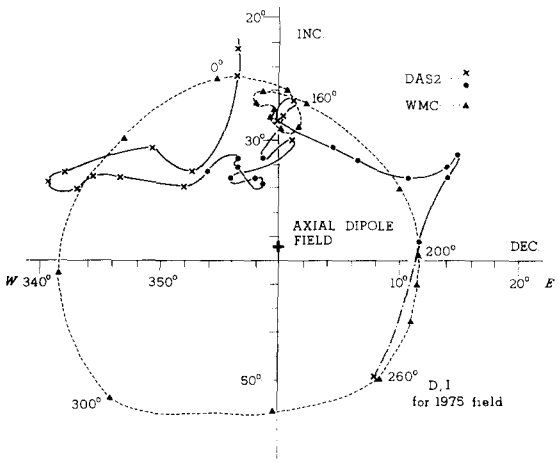


Fig. 8. Bauer plot of DAS2 and the WMC construction. The numbers next to the WMC curve denote present east longitudes having those D and I values.

The shallow inclinations of WMC and BD for longitudes 50°E to 310°E (50°W) are due to the northerly part of the large Gulf of Guinea negative anomaly. The BD construction shows that westerly and easterly declinations are almost symmetrical about the maximum shallow inclinations. It has been hypothesized that such changes can be produced by a westward-drifting radial dipole or current loop at the core-mantle boundary [2,4,23]. This small dipole, for the present-day position of the anomaly, would be situated at about longitude 0°E and latitude 0°N. The analysis of Nevanlinna [23] based on the data of Yukutake and Tachinaka [12] shows that from 1600 AD to 1700 AD the intensity of the anomaly had decreased from −15,000 nT to about −7500 nT, but by 1965 it had grown back again to about −15,000 nT. In this time the focus of the anomaly had maintained constant latitude just north of the geographic equator. (Thompson [24] interprets the data as showing the growth and westerly displacement of one anomaly over the decay of an earlier one. They would thus have shorter lifetimes, and might be considered due to pulsating core sources rather than to drifting ones. The DAS2 data do not preclude such an interpretation.)

In view of the westerly declination feature seen in the stalagmite record, can we deduce the early existence of this anomaly over a thousand years ago? Clearly the recorded inclination changes from 750 to 1000 AD are quite different from those in the WMC or BD constructions. (One possibility is that there is a shallow inclination error in DAS2, although the topographic test appears to rule this out (section 3).)

In Fig. 9, to the DAS2 record, we have added the centred dipole field for Mexico; from worldwide observatory data, as estimated by Barraclough [25], and the 100 year averages from archeomagnetic data, as estimated by Merrill and McElhinny [22]. From about 600 to 1250 AD the main field inclinations were shallow compared to I_{agd} (the estimate of Barbetti [26] over 950–1450 AD also shows this—his mean is at about 1.5°W and $I = 33.5°$), and they would bias the site inclinations to shallower values. However, the fast

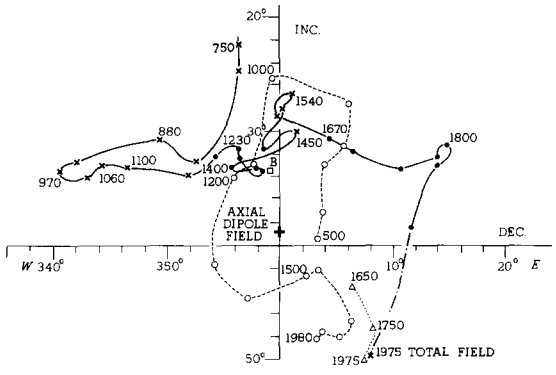


Fig. 9. Bauer plot of DAS2 with the main dipole field D and I values from Merrill and McElhinny [22] (circles and dashed line), of Barraclough [25] (triangles and dotted line), and from Barbetti, from 950 to 1450 AD [26] (square).

change from steep to shallow inclinations, which would accompany the passage of the anomaly, has not been recorded. The MDF inclinations about that time were shallowing (Fig. 7b) and so they would probably not have prevented such an inclination waveform from being recorded. If the Gulf Of Guinea anomaly did exist so as to have drifted westwards past the site about 1000 years ago, then it may be that other proximate non-dipole perturbations changed its inclination signature. Alternatively another explanation must be sought for the persistence of the westerly declination feature.

The interpretation made from these comparisons cannot be rigorous. We would be happier if we had a second Mexican record which would qualify the age data for the earlier part of the DAS2 record. Parts of the DAS2 record have relatively low precision and therefore D-I or VGP segments of the path are not well defined.

9. Conclusions

From these studies and discussions we draw the following conclusions:

(1) A 72 cm high stalagmite, DAS2, from Mexico has been dated by U-Th, ^{14}C and ^{226}Ra methods; after correction for detrital thorium it was found to have grown from 1200 years ago to present.

(2) The stalagmite has yielded a paleomagnetic record of secular variation for this period which is better defined in some parts than in others.

(3) The NRM is derived from the deposition, by cave floods, of magnetite grains on the stalagmite growth surfaces. Chemical effects may have played a part in producing the stable remanence. A topographic test which compares the magnetization of the vertical sides with the horizontal dripcap, did not detect any errors in D or I due to the mode of deposition of the grains.

Comparison of secular variation diagrams has been made with: (a) the present field, at the Mexican latitude, as though it had drifted westwards past the site, (b) the non-dipole field of Bullard et al. [2] as though it had drifted westwards, (c) the movement of the main dipole from 500 AD to present, as estimated by Merrill and McElhinny [22],

These lead us to the following observations and inferences;

(4) A prominent feature of the declinations, near the start of the record, is a large 20° swing to the west. This feature bears great similarity to the declination variations that would result if the Gulf of Guinea negative anomaly had drifted westwards past Mexico about 1000 years ago.

(5) The inclinations of the "westward-drifted" Gulf of Guinea anomaly do not match up so well with those of the early part of the record. The shallow inclination bias may be accounted for if the main dipole was tilted away from the site at these earlier times, as is indicated by the main dipole field averages of Merrill and McElhinny, but the inclination waveform of the anomaly is not recorded.

(6) From the U-Th age data the average drift rate of the non-dipole field at this latitude was estimated to be $0.30 \pm 0.07°\ yr^{-1}$, which compares with an estimate of $0.28°\ yr^{-1}$ by Bullard et al. [2], from modern observatory data. ^{14}C ages from comparable archeomagnetic and lacustrine work also suggest estimates between 0.30 and $0.33°\ yr^{-1}$, though for higher latitudes.

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Appendix 1—Magnetization directions of DAS2

Notes:

(1) All central specimens were AF demagnetized to 55 Oe.

(2) (NS) signifies that the mean is not significant at the 95% confidence level.

(3) N is quoted where one or more specimen directions have been omitted from the computation of the AF mean for that cut, and the data are used in Fig. 3 of the main text.

Cut No.	Distance from top (cm)	NRM					AF demagnetized								
		N	D	I	α_{95}	K	N	D	I	α_{95}	K	δD	δI	VGP	
														lat.	long.
1	1.2	4	10.57	32.43	4.0	312	4	12.87	38.76	3.2	493	4.0	3.2	78.07	−10.38
2	3.1	4	7.39	39.18	11	41	4	10.38	37.94	6.5	115	8.3	6.5	80.32	−6.23
3	5.2	4	24.17	40.88	18	16	4	43.20	39.89	19	14	25	19	50.20	−18.38
4	7.1	4	22.99	15.05	44	2.5(NS)	4	15.90	30.26	6.0	138	7.0	6.0	73.90	9.45
5	9.1	4	337.34	33.86	41	2.9(NS)	4	338.11	49.81	33.2	4	52	33	68.66	−160.85
							(2)	20.41	39.61	12.7	61	16	13	71.12	−3.76
6	11.1	4	20.65	31.89	23	9.5	4	14.80	30.32	13	31	15	13	74.88	10.93
7	13.1	4	10.55	25.12	17	17	4	6.49	29.58	8.7	65	10.0	8.7	81.25	35.40
8	15.1	4	11.05	29.50	12	34	4	6.08	28.12	15.1	22	17	15	90.83	41.03
							(3)	13.34	33.65	5.4	227	6.4	5.4	76.96	5.01
9	17.1	4	40.88	3.69	23	9.2	4	22.33	21.42	8.8	64	9.4	8.8	65.99	14.62
10	19.1	4	23.35	38.73	9.2	58	4	18.76	34.16	8.1	75	9.8	8.1	72.12	−1.61
							(3)	13.84	34.84	5.7	203	6.9	5.7	76.73	1.19
11	21.1	4	7.62	31.85	5.7	153	4	6.28	31.63	3.2	487	3.7	3.2	82.27	29.96
12	23.1	6	7.44	31.81	4.5	161	6	2.99	30.10	4.2	186	4.8	4.2	83.44	55.01
13	25.1	6	3.16	29.11	2.1	735	6	359.83	28.54	2.1	726	2.4	2.1	83.12	82.37
14	27.1	6	2.21	27.58	2.3	609	6	1.03	26.76	2.2	646	2.5	2.2	82.00	73.80
15	29.1	6	7.77	30.03	7.9	52	6	3.40	25.99	7.4	60	8.2	7.4	81.01	59.37
							(5)	359.94	28.41	3.4	333	3.9	3.4	83.04	89.48
16	31.0	6	6.69	31.82	3.6	248	6	2.74	29.36	4.3	173	5.0	4.3	83.11	58.43
							(5)	0.81	30.00	3.7	293	4.2	3.7	83.96	73.58
17	33.0	6	3.21	32.90	6.3	82	6	358.19	31.37	5.5	107	6.5	5.5	84.59	99.68
							(5)	355.73	33.43	2.7	539	3.2	2.7	84.46	128.09
18	35.2	6	2.94	29.52	2.3	612	6	1.39	30.05	2.7	454	3.1	2.7	83.90	68.33
19	37.4	4	358.27	36.35	6.2	126	4	354.67	36.31	5.1	189	6.3	5.1	84.67	150.90
20	39.1	6	9.08	39.86	8.8	42	6	359.30	36.32	10.5	30	13.0	10.5	87.98	100.01
							(5)	356.51	31.20	4.1	232	4.8	4.1	83.81	113.71
21	41.0	6	8.10	28.83	24	5.9	6	356.76	24.43	27	4.6	29	27	80.21	99.91
							(5)	0.73	35.54	7.1	78	8.7	7.1	87.48	90.99
22	43.1	5	0.36	30.20	5.1	151	5	356.54	32.94	5.6	127	6.6	5.6	84.74	119.75
23	45.1	4	5.74	29.84	7.1	98	4	347.72	31.94	7.1	98	8.3	7.1	84.76	105.59
24	47.1	4	355.77	30.63	6.9	102	4	350.25	29.74	9.7	52	11.2	9.7	78.93	138.97
							(3)	349.29	33.91	9.1	79	10.9	9.1	74.37	153.73
25	49.1	4	6.83	38.06	7.3	91	4	356.74	31.01	8.2	72	9.6	8.2	83.83	111.39
26	51.1	4	28.18	50.26	23	9.6	4	355.44	36.90	19	13	24.2	19.4	85.49	152.23
27	53.1	4	353.08	30.67	5.4	166	4	351.80	33.87	3.5	408	4.2	3.5	81.54	147.76
28	55.1	6	350.44	35.65	2.8	415	6	346.64	33.13	2.3	623	2.7	2.3	76.83	155.63
29	57.1	6	347.84	35.37	3.3	298	6	34.20	33.08	2.8	403	3.4	2.8	74.63	158.56
30	59.1	6	347.52	36.76	3.6	248	6	342.81	34.21	2.5	534	3.0	2.5	73.56	162.48

APPENDIX 1 (continued)

Cut No.	Distance from top (cm)	NRM					AF demagnetized								
		N	D	I	α_{95}	K	N	D	I	α_{95}	K	δD	δI	VGP	
														lat.	long.
31	61.1	6	344.13	34.94	2.4	550	6	340.55	33.45	2.4	547	2.9	2.4	71.37	162.71
32	63.1	6	343.34	33.53	2.1	766	6	341.94	32.87	1.0	3068	1.2	1.0	72.53	160.37
33	65.1	6	351.47	30.25	2.8	431	6	349.31	30.59	1.8	1036	2.1	1.8	78.45	143.72
34	67.1	4	0.15	33.34	6.5	115	4	353.23	32.88	3.4	423	4.1	3.4	82.39	138.90
35	69.1	4	3.33	24.34	7.4	88	4	356.34	24.59	3.8	333	4.2	3.8	80.16	102.35
36	70.8	2	7.74	29.47	11	76	2	356.38	22.37	6.6	223	7.2	6.6	78.98	99.88
Lateral specimens															
C30P	59.1		346.55	53.12				348.30	35.81	thermally demagnetized at 200 °C					
C30T	59.1		346.36	56.63				346.38	32.11						

References

1 T. Nagata, Main characteristics of recent geomagnetic secular variation, J. Geomagn. Geoelectr. 17, 263–276, 1965.

2 E.C. Bullard, C. Freedman, H. Gellman and J. Nixon, The westward drift of the Earth's magnetic field, Philos. Trans. R. Soc, London, Ser. A 243, 67–92, 1950.

3 S.K. Runcorn, On the theory of the geomagnetic secular variation, Ann. Geophys. 15, 87–92, 1959.

4 D.D. Skiles, A method of inferring the direction of drift of the geomagnetic field from paleomagnetic data, J. Geomagn. Geoelectr. 22, 441–462, 1970.

5 R.E. Dodson, Counterclockwise precession of the geomagnetic field vector and westward drift of the non-dipole field, J. Geophys. Res. 84, 637–644, 1979.

6 K.M. Creer and P. Tucholka, Secular variation as recorded in lake sediments: a discussion of North American and European results, Philos. Trans. R. Soc. London, Ser. A 306, 87–102, 1982.

7 M.J. Aitken, Physics and Archaeology, 291 pp., Oxford University Press 1974.

8 M. Kovacheva, Archaeomagnetic investigations of geomagnetic secular variations, Philos. Trans. R. Soc. London, Ser. A 306, 79–86, 1982.

9 G.M. Turner and R. Thompson, Lake sediment record of the geomagnetic secular variation in Britain during Holocene times, Geophys. J.R. Astron. Soc. 65, 703–725, 1981.

10 K.M. Creer, D.L. Gross and J.A. Lineback, Origin of regional geomagnetic variations recorded by Wisconsinan and Holocene sediments from Lake Michigan, U.S.A. and Lake Windermere, England, Geol. Soc. Am. Bull. 87, 531–540, 1976.

11 K.M. Creer, Geomagnetic secular variations during the last 25,000 years; and interpretation of data obtained from rapidly deposited sediments, Geophys. J.R. Astron. Soc. 48, 91–109, 1977.

12 T. Yukutake and H. Tachinaka, Separation of the Earth's magnetic field into the drifting and the standing parts, Bull. Earthq. Res. Inst. Tokyo Univ. 47, 65–97, 1969.

13 T. Yukutake, Review of the geomagnetic secular variations of the historical time scale, Phys. Earth Planet. Inter. 20, 83–95, 1979.

14 M.W. McElhinny and W.E. Senanayake, Variations in the geomagnetic dipole 1: the past 50,000 years, J. Geomagn. Geoelectr. 17, 39–51, 1982.

15 A.G. Latham, H.P. Schwarcz, D.C. Ford and G.W. Pearce, Palaeomagnetism of stalagmite deposits, Nature 280, 383–385, 1979.

16 A.G. Latham, H.P. Schwarcz, D.C. Ford and G.W. Pearce, The paleomagnetism and U-Th dating of three Canadian speleothems: evidence for the westward drift, 5.4–2.1 ka BP, Can. J. Earth Sci. 19, 1985–1995, 1982.

17 A. Kaufman and W.S. Broecker, Comparison of ^{230}Th and ^{14}C ages for carbonate materials from Lakes Lahontan and Bonneville. J. Geophys. Res. 70, 4039–4034, 1965.

18 T-H. Peng, J.G. Goddard and W.S. Broecker, A direct comparison of ^{14}C and ^{230}Th ages at Searle Lake, California, Quat. Res. 9, 319–329, 1978.

19 M. Gascoyne, Pleistocene climates determined from stable isotope and geochronologic studies of speleothems, Ph.D. Thesis, McMaster University, 1979.

20 R. Sternberg, Archaeomagnetism in the Southwest of North America, in, Geomagnetism of Baked Clays and Recent Sediments, K.M. Creer, P. Tucholka and C.E. Barton, eds., 324 pp., Elsevier, 1983.

21 K.L. Verosub and P.J. Mehringer, Congruent paleomagnetic and archeomagnetic records from the western United States: AD 750 to 1450, Science 224, 387–389, 1984.

22 R.T. Merrill and M.W. McElhinny, The Earth's Magnetic Field, 401 pp., Academic Press, 1983.

23 H. Nevanlinna, Interpreting regional geomagnetic anomalies and their secular variation with the aid of radial dipoles and current loops, J. Geomagn. Geoelectr. 32, 483–506, 1980.

24 R. Thompson, Geomagnetic evolution: 400 years of change on planet Earth, Phys. Earth Planet. Inter. 36, 61–77, 1984.

25 D.R. Barraclough, Spherical harmonic analyses of the geomagnetic field for eight epochs between 1600 and 1910, Geophys. J.R. Astron. Soc. 36, 497–513, 1974.

26 M. Barbetti, Measurement of recent geomagnetic secular variation in southeastern Australia and the question of dipole wobble, Earth Planet. Sci. Lett. 36, 207–218, 1977.

27 M. Stuiver and H.A. Polach, Discussion: Reporting of ^{14}C data, Radiocarbon 19, 355–363, 1977.